

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276373

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Nelson; Scott et al.

ADAPTIVE DEPOSITION FOR INITIAL LAYER OF ADDITIVE MANUFACTURING PROCESS

Abstract

An additive manufacturing system includes an energy delivery device to deliver energy to a build surface of a deposit overlying a substrate to form a melt pool in the build surface, a powder delivery device to direct a powder stream toward the melt pool, and a computing device to determine a first set of deposition parameters for an innermost layer of the deposit overlying the substrate, determine a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer, determine a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers, and control the energy delivery device and the powder delivery device to deposit the innermost layer, the inner plurality of layers, and the outer plurality of layers based on the respective first, second, and third sets of deposition parameters.

Inventors: Nelson; Scott (Indianapolis, IN), Puhl; David James (Indianapolis, IN), Grafton-Reed; Clive (London, GB), Daum; Peter E. (Indianapolis, IN), Proctor; Robert F. (Indianapolis, IN), Heason; Christopher P. (London, GB)

Applicant: Rolls-Royce Corporation (Indianapolis, IN); Rolls-Royce plc (London, GB)

Family ID: 96881588

Appl. No.: 18/593235

Filed: March 01, 2024

Publication Classification

Int. Cl.: B22F10/85 (20210101); B22F10/368 (20210101); B33Y30/00 (20150101); B33Y50/02 (20150101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to additive manufacturing techniques.

BACKGROUND

[0002] Additive manufacturing generates three-dimensional structures through addition of material layer-by-layer or volume-by-volume to form the structure, rather than removing material from an existing component to generate the three-dimensional structure. Additive manufacturing may be advantageous in many situations, such as rapid prototyping, forming components with complex three-dimensional structures, or the like. In some examples, additive manufacturing may utilize powdered materials and may melt or sinter the powdered material together in predetermined shapes to form the three-dimensional structures. In some instances, the three-dimensional structures may be formed on an underlying substrate, resulting in an interface that may be subject to different deposition conditions or operating stresses than other portions of the three-dimensional structures.

SUMMARY

[0003] The disclosure describes additive manufacturing systems, and methods for operating additive manufacturing systems, that manufacture a component by carefully controlling deposition of layers close to an underlying substrate. Layers of a deposit close to the underlying substrate may have a greater impact on adhesion of the deposit and integrity of the deposit and substrate than layers further from the substrate. The additive manufacturing system uses one or more sets of deposition parameters for these close layers that are different from layers further away from the substrate. For example, a layer directly overlying the substrate may be deposited in a manner that limits a temperature of the substrate and increases adhesion of the layer to the substrate. As another example, various inner layers overlying the substrate may be deposited in a manner that avoids melt back of the substrate and reduces defects in the inner layers. The additive manufacturing system may use in-situ monitoring of the melt pool to further control the deposition parameters to avoid damage that may be indicated by a temperature or size of the melt pool. In this way, a resulting component may have reduced damage to the substrate, reduced damage to the deposit, and/or greater adhesion between the deposit and the substrate compared to a component in which a same or similar set of deposition parameters are used to deposit layers close to and distant from the substrate.

[0004] In some examples, the disclosure describes an additive manufacturing system that includes an energy delivery device, a powder delivery device, and a computing device. The energy delivery device is configured to deliver energy to a build surface of a deposit overlying a substrate to form a melt pool in the build surface. The powder delivery device is configured to direct a powder stream toward the melt pool. The computing device is configured to determine a first set of deposition parameters for an innermost layer of the deposit overlying the substrate, determine a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer, and determine a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers, in which the third set of deposition parameters is different from the first and second sets of deposition parameters. The computing device is further configured to control the energy delivery device and the powder delivery device to deposit the innermost layer on the substrate based on the first set of deposition parameters, deposit the inner plurality of layers on the innermost layer based on the second set of deposition parameters, and deposit the outer plurality of layers on the inner plurality of layers based on third set of deposition parameters. When

operating based on the first set of deposition parameters, the energy delivery device delivers a first amount of energy to the build surface. When operating based on the second set of deposition parameters, the energy delivery device delivers a second amount of energy to the build surface. When operating based on the third set of deposition parameters, the energy delivery device delivers a third amount of energy to the build surface, which is different from each of the first and second amounts of energy.

[0005] In some examples, the disclosure describes a method for additive manufacturing that includes determining, by a computing device, a first set of deposition parameters for an innermost layer of a deposit overlying a substrate. The method further includes determining, by the computing device, a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer. The method further includes determining, by the computing device, a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers. The third set of deposition parameters is different from the first and second sets of deposition parameters. The method further includes controlling, by the computing device, an energy delivery device configured to deliver energy to the build surface and a powder delivery device configured to direct a powder stream toward the melt pool to deposit the innermost layer on the substrate based on the first set of deposition parameters, deposit the inner plurality of layers on the innermost layer based on the second set of deposition parameters, and deposit the outer plurality of layers on the inner plurality of layers based on third set of deposition parameters. When operating based on the first set of deposition parameters, the energy delivery device delivers a first amount of energy to the build surface. When operating based on the second set of deposition parameters, the energy delivery device delivers a second amount of energy to the build surface. When operating based on the third set of deposition parameters, the energy delivery device delivers a third amount of energy to the build surface, which is different than each of the first and second amounts of energy.

[0006] The details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] FIG. 1A is a conceptual block diagram illustrating mass flow monitoring aspects of an example additive manufacturing system that includes a powder flow monitoring system and a topology sensor for monitoring mass flow of powder during an additive manufacturing technique.

[0008] FIG. 1B is a conceptual block diagram illustrating heat flow monitoring aspects of the example additive manufacturing system of FIG. 1A, including a melt pool monitoring system, an optical system, and a thermal sensor for monitoring a melt pool during an additive manufacturing technique.

[0009] FIG. 2 is a process flow diagram illustrating a mass flux and heat flux monitoring and control technique.

[0010] FIG. 3 is a flowchart illustrating an example method for fabricating a component.

[0011] FIG. 4A is a cross-sectional side view diagram illustrating deposition of a first layer on a substrate.

[0012] FIG. 4B is a cross-sectional side view diagram illustrating deposition of an inner plurality of layers on the first layer.

[0013] FIG. 4C is a cross-sectional side view diagram illustrating deposition of an outer plurality of layers on the inner plurality of layers.

DETAILED DESCRIPTION

[0014] The disclosure generally describes techniques and systems for fabricating a component using a blown powder additive manufacturing technique, such as a directed energy deposition (DED) technique, by carefully controlling deposition of layers close to an underlying substrate. During blown powder additive manufacturing, a component is built up by adding material to the component in sequential layers. The final component is composed of a plurality of layers of material. In some blown powder additive manufacturing techniques for forming components from metals or alloys, an energy source may direct energy at a substrate to form a melt pool. A powder delivery device may deliver a powder to the melt pool, where at least some of the powder at least partially melts and is joined to the melt pool and, thus, substrate. The properties of the final component, including the presence or absence of material defects and the adhesion of the deposit to the substrate, are a function of a number of variables related to mass flux and heat flux.

[0015] Layers of a deposit close to the underlying substrate may have a greater impact on adhesion of the deposit and integrity of the deposit and substrate than layers further from the substrate. For example, the deposit and the substrate may be formed from different materials that have different properties, including different thermal properties such as melting points or coefficients of thermal expansion. During deposition of an innermost layer of a deposit, a surface portion of the substrate may be partially melted to mix with a portion of the innermost layer. While this melting may adhere the innermost layer to the substrate, underlying portions of the substrate may be exposed to high temperatures, damaging the substrate. Subsequent layers deposited on the innermost layer and close to the substrate may further expose the substrate to temperatures that cause the substrate to remelt, which may reduce an adhesion between the substrate and the innermost layer. Further, the innermost or subsequent layers may be subject to thermal stresses between the substrate and the deposit caused by a change in temperature, which may cause any defects or other anomalies in these layers to undergo damage, such as in the form of cracks.

[0016] In accordance with techniques of this disclosure, an additive manufacturing system may use one or more sets of deposition parameters for these close layers that are different from layers further away from the substrate. For example, the inner layer may be deposited in a manner that limits a temperature of the substrate and increases adhesion of the layer to the substrate. The various subsequent layers overlying the inner layer may be deposited in a manner that avoids melt back of the substrate and reduces defects in these layers. The additive manufacturing system may use in-situ monitoring to further control the deposition parameters to avoid damage to the substrate that may be indicated by a temperature or size of the melt pool. In this way, a resulting component may have reduced damage to the substrate, reduced damage to the deposit, and/or greater adhesion between the deposit and the substrate compared to a component in which a same or similar set of deposition parameters are used to deposit layers close to and distant from the substrate.

[0017] FIG. 1A is a conceptual block diagram illustrating mass flow monitoring aspects of an example additive manufacturing system **10** that includes a powder flow monitoring system (PFMS) **18** and a topology sensor **48** for monitoring mass flow of powder during an additive manufacturing technique. In the example illustrated in FIG. 1A, additive manufacturing system **10** includes a computing device **12**, a powder delivery device **14**, an energy delivery device **16**, PFMS **18**, a stage **20**, a powder source **42**, powder source mass sensor **44**, and topology sensor **48**. Computing device **12** is operably connected to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and topology sensor **48**. To simplify illustration of FIG. 1A and improve clarity of the figure, heat flow monitoring aspects of additive manufacturing system **10** are shown in FIG. 1B and described below with reference to FIG. 1B.

[0018] In some examples, stage **20** is movable relative to energy delivery device **16** and/or energy delivery device **16** is movable relative to stage **20**. Similarly, stage **20** may be movable relative to powder delivery device **14** and/or powder delivery device **14** may be movable relative to stage **20**. For example, stage **20** may be translatable and/or rotatable along at least one axis to position component **22** relative to energy delivery device **16** and/or powder delivery device **14**. Similarly,

energy delivery device **16** and/or powder delivery device **14** may be translatable and/or rotatable along at least one axis to position energy delivery device **16** and/or powder delivery device **14**, respectively, relative to component **22**. Stage **20** may be configured to selectively position and restrain component **22** in place relative to stage **20** during manufacturing of component **22**.

[0019] Powder source **42** is the source of powder for powder stream **30**. Powder source **42** may include any suitable container or enclosure, such as a hopper, configured to hold powder. Powder source **42** also may include mechanism for entraining the powder in a gas flow. For instance, powder source **42** may be coupled to a gas source, which provides a gas flowing through powder source **42** and entraining powder within the gas flow. Additionally, or alternatively, powder source **42** may include an agitator configured to agitate the powder and increase entrainment of the powder in the gas stream. System **10** may include a powder source mass sensor **44** associated with powder source **42**. Powder source mass sensor **44** may be configured to quantify loss of mass in the powder source **42** or, alternatively, a mass flow out of powder source **42**.

[0020] Powder source **42** is fluidically coupled to powder delivery device **14** via a flow path **46**. Flow path **46** may include any suitable structure(s) defining an enclosed flow between powder source **42** and powder delivery device, including conduit, pipe, tubes, or the like. Although not shown in FIG. **1A**, for at least part of flow path **46** between powder source **42** and nozzles of powder delivery device **14**, flow path **46** may split into multiple, parallel sections, e.g., one for each nozzle. Further, although not shown in FIG. **1A**, in some examples, flow path **46** may include one or more nozzles for controlling flow through flow path **46** as a whole or portions of flow path **46** (e.g., a section associated with a particular nozzle of powder delivery device **14**).

[0021] Powder delivery device **14** may be configured to deliver powder to selected locations of component **22** being formed via a powder stream **30**. Powder delivery device **14** may include one or more nozzles that each output powder. The combined powder defines powder stream **30**. In some examples, powder delivery device **14** includes a single nozzle, which may be point nozzle, or a single nozzle that is an annular channel. In other examples, powder delivery device **14** includes a plurality of nozzles (e.g., three nozzles or four nozzles). Regardless of the number of nozzles, powder delivery device **14** may output a powder stream that is focused at a focus plane. As powder delivery device **14** is movable in the z-axis shown in FIG. **1A** relative to component **22**, the focal plane of powder delivery device **14** also may be movable in the z-axis relative to component **22**, such that the focus plane may be controlled to be substantially coincident with build surface **28**. At least some of the powder in powder stream **30** may impact a melt pool **32** in component **22**. At least some of the powder that impacts melt pool **32** may be joined to component **22**.

[0022] Energy delivery device **16** may include an energy source, such as a laser source, an electron beam source, plasma source, or another source of energy that may be absorbed by component **22** to form a melt pool **32** and/or be absorbed by powder in powder stream **30** to be added to component **22**. Example laser sources include a CO laser, a CO.sub.2 laser, a Nd:YAG laser, or the like. In some examples, the energy source may be selected to provide energy with a predetermined wavelength or wavelength spectrum that may be absorbed by component **22** and/or the powder to be added to component **22** during the additive manufacturing technique. In some examples, energy delivery device **16** also includes an energy delivery head, which is operatively connected to the energy source. The energy delivery head may aim, focus, or direct energy **34** toward predetermined positions at or adjacent to a surface of component **22** during the additive manufacturing technique. As described above, in some examples, the energy delivery head may be movable in at least one dimension (e.g., translatable and/or rotatable) under control of computing device **12** to direct the energy toward a selected location at or adjacent to a surface of component **22**.

[0023] In some examples, at least a portion of energy delivery device **16** and powder delivery device **14** may be combined or attached to each other. For example, a deposition head may include part of powder delivery device **14** (e.g., internal channels and powder nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**) and part of energy delivery

device **16** (e.g., the energy delivery head). As shown in FIG. **1A**, in some examples, energy delivery device **16** may be arranged or configured such that energy **34** and powder stream **30** both exit from a common deposition head and are directed toward build surface **28**. For instance, energy **34** may pass through a central channel within the deposition head and exit a central aperture in the deposition head, while fluidized powder may flow through internal channels and powder nozzle(s) for forming powder stream **30** and directing powder stream **30** toward build surface **28**.

[0024] System **10** includes powder flow monitoring system (PFMS) **18**. PFMS **18** is configured to image at least a portion of powder stream **30** to detect powder flowing between powder delivery device **14** and build surface **28**. For example, PFMS **18** may include an illumination device and an imaging device. In some examples, the illumination device may include one or more light source. For instance, the illumination device may include one or more structured light devices, such as one or more lasers. The illumination device is configured to illuminate a plane of powder stream **30** at image plane **38**, e.g., a plane substantially perpendicular to an axis extending between powder delivery device **14** and build surface **28**. The imaging device of PFMS **18** is configured to image at least some of the illuminated powder. The imaging device may have a relatively high data acquisition speed (e.g., frame rate), such greater than 1000 Hz. Because of the velocity of the powder in powder stream **30**, even such a frame rate may image only a fraction of the powder flowing between powder delivery device **14** and build surface **28**.

[0025] PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16**, e.g., in the x-y plane shown in FIG. **1A**. PFMS **18** may be movable in the z-axis direction of FIG. **1A** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). This may enable movement of image plane **38** along the z-axis of FIG. **1A** (e.g., parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28**). In this way, PFMS **18** may analyze powder stream **30** along powder stream **30** to help determine parameters of powder stream **30** along its length.

[0026] In some example, PFMS **18** may be positionally fixed relative to powder delivery device **14** and/or energy delivery device **16** and movable parallel to a longitudinal axis extending from powder delivery device **14** to build surface **28** by an adjustable z-stage **40**. Adjustable z-stage **40** may be attached to energy delivery device **16**, powder delivery device **14**, or a portion of system **10** that moves energy delivery device **16** and/or powder delivery device **14**, such that PFMS **18** moves in the x-y axis in registration with energy delivery device **16** and/or powder delivery device **14**. Adjustable z-stage **40** may be controlled by computing device **12** to position PFMS **18** and image plane **38** relative to powder stream **30**. Further, computing device **12** may control adjustable z-stage **40** to move PFMS **18** vertically and out of the way to allow powder delivery device **14** and energy delivery device **16** access to physically constrained areas, e.g., between vanes of a doublet or triplet of a nozzle guide vane for a gas turbine engine.

[0027] System **10** further includes a topology sensor **48**. Topology sensor **48** is configured to monitor an amount of powder captured by melt pool **32** by imaging melt pool **32** and the added material, allowing the mass to be quantified (e.g., by computing device **12**) using the dimensions of the added material and density of the material (powder). In some examples, topology sensor **48** includes a laser and a sensor (e.g., an imaging device), which senses laser light reflected by the structure being imaged (e.g., melt pool **32** and the added material). The laser may have a defined wavelength, which may affect the resolution of the topology sensor **48**. In some examples, the wavelength and sensor may be selected such that the resolution of topology sensor **48** is as great as about 10 microns (e.g., about 6 microns). In some examples, topology sensor **48** may be positioned substantially directly above component **22** and may include an interferometer, which provides depth information based on the time from outputting a laser pulse to the sensing of the reflected light. In other examples, topology sensor **48** may be positioned at an offset with respect to component **22** such that the sensor senses depth information without using an interferometer.

[0028] Computing device **12** is configured to control components of system **10** and may include,

for example, a desktop computer, a laptop computer, a workstation, a server, a mainframe, a cloud computing system, or the like. Computing device **12** is configured to control operation of system **10**, including, for example, powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48**. Computing device **12** may be communicatively coupled to powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** using respective communication connections. In some examples, the communication connections may include network links, such as Ethernet, ATM, or other network connections. Such connections may be wireless and/or wired connections. In other examples, the communication connections may include other types of device connections, such as USB, IEEE 1394, or the like. [0029] Although FIG. **1A** illustrates a single computing device **12** and attributes all control and processing functions to that single computing device **12**, in other examples, system **10** may include multiple computing devices **12**, e.g., a plurality of computing devices **12**. In general, control and processing functions described herein may be divided among one or more computing devices. For instance, system **10** may include controller for energy delivery device **16**, powder delivery device **14**, and stage **20**, a separate controller for PFMS **18**, and a separate computing device for analyzing data obtained by PFMS **18**, mass sensor **44**, and topology sensor **48**. As another example, system may include a dedicated controller for each of energy delivery device **16**, powder delivery device **14**, stage **20**, PFMS **18**, and topology sensor **48**, and a separate computing device for coordinating control of powder delivery device **14**, energy delivery device **16**, PFMS **18**, stage **20**, powder source **42**, powder source mass sensor **44**, and/or topology sensor **48** and analyzing data obtained by PFMS **18** powder source mass sensor **44**, and/or topology sensor **48**. Other examples of computing system architectures for controlling system **10** and analyzing data obtained from system **10** will be apparent and are within the scope of this disclosure.

[0030] Computing device **12** may be configured to control operation of powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40**, stage **20**, and/or topology sensor **48** to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**. For example, as described above, computing device **12** may control stage **20** and powder delivery device **14**, energy delivery device **16**, adjustable z-stage **40** and/or topology sensor to translate and/or rotate along at least one axis to position component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**. Positioning component **22** relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48** may include positioning a predetermined surface (e.g., a surface to which material is to be added) of component **22** in a predetermined orientation relative to powder delivery device **14**, energy delivery device **16**, PFMS **18**, and/or topology sensor **48**.

[0031] Computing device **12** may be configured to control system **10** to deposit layers **26A** and **26B** on a substrate **24** to form component **22** based on a set of deposition parameters **13**. As shown in FIG. **1A**, component **22** may include a first layer **26A** overlying substrate **24** and a second layer **26B** overlying first layer **26A**, although many components may be formed of additional layers, such as tens of layers, hundreds of layers, thousands of layers, or the like. Component **22** in FIG. **1A** is simplified in geometry and the number of layers compared to many components formed using additive manufacturing techniques. Although techniques are described herein with respect to component **22** including substrate **24**, first layer **26A** and second layer **26B**, the technique may be extended to components **22** with more complex geometry and any number of layers.

[0032] To form component **22**, computing device **12** may control powder delivery device **14** and energy delivery device **16** to form, on a surface **28** of first layer **26A** of material, a second layer **26B** of material using an additive manufacturing technique. Computing device **12** may control energy delivery device **16** to deliver energy **34** to a volume at or near surface **28** to form melt pool **32**. For example, computing device **12** may control the relative position of energy delivery device **16** and stage **20** to direct energy to the volume. Computing device **12** also may control powder

delivery device **14** to deliver powder stream **30** to melt pool **32**. For example, computing device **12** may control the relative position of powder delivery device **14** and stage **20** to direct powder stream **30** at or on to melt pool **32**. Computing device **12** may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26B** is complete. Computing device **12** then may control a z-axis position of stage **20** and/or powder delivery device **14** and energy delivery device **16** such that melt pool **32** will be formed on surface **36** of second layer **26B**, and may control powder delivery device **14** and energy delivery device **16** to move energy **34** and powder stream **30** along build surface **28** in a pattern until layer **26B** is complete. Computing device **12** may control powder delivery device **14** and energy delivery device **16** similarly until all layers are formed to define a completed component **22**.

[0033] As described above, system **10** may include both mass flow monitoring and heat flow monitoring, but FIG. **1A** illustrates only mass flow monitoring aspects of system **10**. FIG. **1B** is a conceptual block diagram illustrating heat flow monitoring aspects of the example additive manufacturing system of FIG. **1A**, including a melt pool monitoring system (MPMS) **15**, an optical system **80**, and a thermal sensor **82** for monitoring a melt pool, such as a size and/or temperature, during an additive manufacturing technique. Identical reference numerals in FIGS. **1A** and **1B** refer to the same parts. Further, those common parts are the same or substantially identical, aside from any differences described herein.

[0034] As shown in FIG. **1B**, system **10** includes an optical system **80**. During additive manufacturing, component **22** is built up by adding material to component **22** in sequential layers. The final component is composed of a plurality of layers of material. Energy delivery device **16** may direct energy **34** at first layer **26A** to form melt pool **32**. Powder delivery device **14** may deliver powder stream **30** to melt pool **32**, where at least some of the powder at least partially melts and is joined to first layer **26A**. Melt pool **32** cools as energy **34** is no longer delivered to that location of first layer **26A** (e.g., due to energy delivery device **16** scanning energy **34** over the surface of first layer **26A**). The temperature and cooling rate of melt pool **32** and the surrounding areas of first layer **26A** affect the microstructure of the component **22** formed using the additive manufacturing technique.

[0035] In many cases, energy **34** output by energy delivery device **16** is very high temperature and the intensity of its thermal emissions is significantly greater than the intensity of thermal emissions from melt pool **32** and the surrounding areas. Similarly, thermal emissions intensity at and near the center of melt pool **32** may be significantly greater than the intensity of thermal emissions near the edge of melt pool **32** and in areas surrounding melt pool **32**. Because of this, it may be difficult to accurately measure temperature and cooling rate of areas near the edge of melt pool **32** and in areas surrounding melt pool **32**. This results in difficulty predicting and controlling microstructure of the additively manufactured component **22**.

[0036] Optical system **80** may include an imaging device and an associated optical train, which senses emissions at or near component **22** during the additive manufacturing technique. For example, optical system **80** may include a visible light imaging device, an infrared imaging device, or an imaging device that is configured (e.g., using a filter) to image a specific wavelength or wavelength range.

[0037] The optical train may include one or more reflective, refractive, diffractive optical components configured to direct light to the imaging device. For example, the optical train may be configured to direct light from near component **22** and/or melt pool **32** to the imaging device. In some examples, at least a portion of the optical train is coaxial with the axis at which energy delivery device **16** outputs energy, and the at least a portion of the optical train may be attached to or otherwise configured to move with the portion of energy delivery device **16** that directs or focuses energy **34** at or near the surface of component **22**. In this way, optical system **80** may move with energy delivery device **16** and track melt pool **32** as melt pool **32** moves across component **22**,

without needing to correct for any offsets between energy delivery device **16** and optical system **80** and/or needing to correct for geometry of component **22**. In other examples, the optical train may not be coaxial with the axis at which energy delivery device **16** outputs energy **34**, and computing device **12** may be configured to compensate for the offset and any affects this may have on the imaging, including shadowing, interference, geometry of component **22**, or the like.

[0038] Optical system **80** may include an occulting device. The occulting device is configured to reduce or block emissions (e.g., thermal emissions) that originate from the energy output by energy delivery device **16** and/or near a center of melt pool **32**, which otherwise obfuscate emissions from solidifying regions of material at or near the edge of melt pool **32** and outside of melt pool **32**. The occulting device may be a rigid occulting device or a dynamic occulting device. A rigid occulting device reduces or blocks emissions from a fixed region, e.g., from the energy **34** output by energy delivery device **16**. For instance, a rigid occulting device may include a device with fixed dimensions that is opaque to wavelengths of interest. As another example, a rigid occulting device may include an apodizing lens in which a center of the lens is substantially opaque to wavelengths of interest and opacity decreases as a function of radius.

[0039] A dynamic occulting device is configured to be controlled to occult different regions, e.g., different sizes and/or shapes. A dynamic occulting device may include a rigid occulting device that is mounted to a device that can translate the rigid occulting device along and/or perpendicularly to the optical axis. As another example, a dynamic occulting device may include an opaque and viscous liquid, such as mercury, contained between two substrates. The substrates are substantially transparent to the wavelength(s) of interest. One or both of the substrates may be movable relative to the other substrate to control the distance between the substrates. By reducing the distance between the substrates, the size of the occulting region may increase. By increasing the distance between the substrates, the size of the occulting region may decrease. As a third example, a dynamic occulting device may include a digital micromirror device. Computing device **12** may be configured to control the micromirrors of the digital micromirror device to direct emissions that originate from energy **34** output by energy delivery device **16** and/or near a center of the melt pool away from the imaging device. A digital micromirror device may enable control of both the size and shape of the region of emissions that are occulted.

[0040] Surrounding melt pool **32** is a cooling zone, in which temperature gradients from the temperature of melt pool **32** to ambient temperature are present. Melt pool **32** and the cooling zone may emit thermal emissions (e.g., thermal radiation), which travel through optical system **80** to an imaging device, which images the thermal emissions. The occulting device may occult (e.g., reduces the intensity of or substantially eliminates) thermal emissions from a selected region, e.g., a region corresponding to energy **34** and at least a portion of melt pool **32**. This may allow the imaging device to more effectively image relatively lower intensity thermal emissions from at or near the edge of melt pool **32** and within the cooling zone. This may enable more accurate measurement of temperatures within the cooling zone, and heat flow within the cooling zone.

[0041] System **10** includes melt pool monitor system (“MPMS”) **15**. MPMS **15** may be communicatively coupled to a thermal sensor **82** for monitoring a characteristic of melt pool **32**. Thermal sensor **82** may include an imaging system, such as a visual or thermal camera, e.g., camera to visible light or infrared (IR) radiation. A visible light camera may monitor the geometry of the melt pool, e.g., a width, diameter, shape, or the like. A thermal (or IR) camera may be used to detect the size, temperature, or both of the melt pool. In some examples, a thermal camera may be used to detect the temperature of the melt pool at multiple positions within the melt pool, such as a leading edge, a center, and a trailing edge of the melt pool. In some examples, the imaging system may include a relatively high speed camera capable of capturing image data at a rate of tens or hundreds of frames per second or more, which may facilitate real-time detection of the characteristic of the melt pool.

[0042] FIG. **2** is a process flow diagram illustrating a mass flux and heat flux monitoring and

control technique. The technique of FIG. 2 may be implemented by system **10** of FIGS. **1A** and **1B** and will be described with concurrent reference to FIGS. **1A** and **1B**. However, it will be appreciated that system **10** may perform other techniques and the technique of FIG. 2 may be performed by other systems.

[0043] One or more computing devices **12** may be configured to control a powder feed rate output by powder source **42** (see top left of FIG. 2). For instance, one or more computing devices **12** may be configured to control an agitator of powder source **42**, a gas flow rate of gas flowing through powder source **42**, a position of one or more valves within flow path **46**, or the like to control a powder feed rate output by powder source **42**. One or more computing devices **12** may be configured to receive data from one or more mass flow monitoring sensors, including PFMS **18**, powder source mass sensor **44**, and/or topology sensor **48**. Data received from powder source mass sensor **44** indicates a mass flow of powder from powder source **42** to powder delivery device. Data from PFMS **18** indicates a mass flow of powder in powder stream **30** between powder delivery device **14** to adjacent melt pool **32**. Data from topology sensor **48** indicates powder mass captured by melt pool **32** and added to component **22**.

[0044] One or more computing devices **12** may calculate one or more mass flow-related metrics based on the data received from PFMS **18**, powder source mass sensor **44**, and/or topology sensor **48**. For example, one or more computing devices **12** may determine a capture efficiency by determining a fraction or percentage of powder from powder stream **30** that is captured by melt pool **32** and added to component **22**, e.g., by dividing the powder mass captured by melt pool **32**, as determined based on data from topology sensor, into the mass flow determined based on data received from PFMS **18**. Further, one or more computing devices **12** may determine an overall mass flux using the data received from PFMS **18**, powder source mass sensor **44**, and/or topology sensor **48**. One or more computing devices **12** then may use the overall mass flux as an input to the control algorithm used to control the powder feed rate output by powder source **42** (see top left of FIG. 2).

[0045] Similarly, one or more computing devices **12** may be configured to control energy delivery device **16** to deliver energy **34** to first layer **26A** to establish a given heat input (see bottom left of FIG. 2). For example, one or more computing device **12** may control one or more operating parameters of energy delivery device **16**, such as intensity, pulse rate, pulse width, or the like; one or more positional parameters related to energy delivery device **16**, such as dwell time at a location, a movement rate relative to first layer **26A**, an overlap between adjacent passes of energy **34** across first layer **26A**, a pause time between adjacent passes of energy **34** across first layer **26A**, or the like to control heat input to system **10** (e.g., to melt pool **32** and component **22**). One or more computing devices **12** may be configured to receive data from one or more heat sensors, such as MPMS **15**, optical system **80**, and/or thermal sensor **82**.

[0046] One or more computing devices **12** may determine a cooling rate and associated heat from using thermal data from thermal sensor **82** and/or optical system **80** and may determine a heat input into component **22** using a size and/or temperature of melt pool **32** as observed by melt pool monitor. One or more computing devices **12** may be configured to determine an overall heat flux using these data. One or more computing devices **12** then may use the overall heat flux as an input to the control algorithm used to control the energy delivery by energy delivery device **16** (see top left of FIG. 2). In some examples, one or more computing devices **12** also may use the deposit topology (captured powder mass) and/or capture efficiency metric in the determination of the heat flux, as the added powder mass and quench effects associated with the captured powder affect the cooling rate.

[0047] FIG. 3 is a flowchart illustrating an example method for fabricating a component. The example method of FIG. 3 will be described with respect to FIGS. 4A-4C below. The example method of FIG. 3 includes determining a first set of deposition parameters for an innermost layer of a deposit overlying a substrate (**100**). FIG. 4A is a cross-sectional side view diagram illustrating

deposition of an innermost layer **124** of a deposit **120A** on a substrate **122**. The first set of deposition parameters is configured to cause an energy delivery device to deliver a first amount of energy **128A** to a build surface **123A** of deposit **120A** to form a melt pool **126A** in build surface **123A**, and cause a powder delivery device to direct a powder stream **129A** toward melt pool **126A**, thereby fusing the powder and forming innermost layer **124**. An amount of energy may be quantified in any number of ways including, but not limited to, a heat flux, a beam power, or any other metric that quantifies energy delivered to a particular area or volume.

[0048] Energy **128A** delivered to build surface **123A** to form melt pool **126A** may cause at least a portion of substrate **122** to melt, as indicated by a melt zone **125**. In addition to melt zone **125**, energy **128A** may further cause a portion of substrate **122** to heat above a temperate that may affect an integrity of substrate **122**, as indicated by heat affected zone **127A**. In examples in which deposit **120A** and substrate **122** are formed from different materials, substrate **122** may be particularly susceptible to thermal damage due to different thermal properties of substrate **122** and deposit **120A**. For example, a melting point and/or sintering temperature of deposit **120A** may be greater than a melting point of substrate **122**.

[0049] The first set of deposition parameters may be configured to maintain a temperature of substrate **122** below a temperature that corresponds to thermal damage of substrate **122**. A lower temperature of substrate **122** may correspond to a reduced amount of thermal damage to substrate **122**. The first set of deposition parameters may reduce a size of heat affected zone **127A** while still maintaining adequate energy **128A** for forming innermost layer **124**. A temperature of substrate **122** and size or depth of heat affected zone **127A** may be influenced by the first amount of energy **128A** input into melt pool **126A** and an amount of energy dissipated from substrate **122**. As such, the first set of deposition condition may include deposition parameters that are configured to reduce an amount of energy **128A** input into melt pool **126A** and/or increase an amount of energy dissipated from portions of substrate **122** proximate to melt pool **126A**.

[0050] In some examples, the first set of deposition parameters may be configured to limit the first amount of energy **128A** to build surface **135A** to reduce a temperature of substrate **122**. For example, the first set of deposition parameters may be configured to maintain a size and/or temperature of melt pool **126A** at or below a threshold that corresponds to thermal damage of substrate **122**. A smaller size and/or lower temperature of melt pool **126A** may generally result in a lower temperature of substrate **122** and, correspondingly, a reduced amount of thermal damage to substrate **122**. The first set of deposition parameters may be configured to reduce a size of heat affected zone **127A** while still maintaining adequate energy **128A** for forming innermost layer **124**.

[0051] In some examples, the first set of deposition parameters may be configured to permit cooling of substrate **122** to reduce a temperature of substrate **122**. For example, the first set of deposition parameters may define a toolpath of melt pool **126A** that permits portions of substrate **122** recently exposed to energy **128A** sufficient time to cool, such that a temperature of the particular portion may be reduced. Deposition parameters related to the toolpath may include, but are not limited to, a length of time of move pauses (e.g., increasing a time of move pauses), a spatial configuration of the toolpath (e.g., increasing a spacing between a particular run), and any other parameter that may affect an amount time between exposures to energy **128A**.

[0052] The example method of FIG. 3 includes depositing innermost layer **124** on substrate **122** based on the first set of deposition parameters (**102**). For example, computing device **12** may control energy delivery device **16** and powder delivery device **14** to deposit innermost layer **124** on substrate **122** based on the first set of deposition parameters. During deposition, a temperature of substrate **122** may remain sufficiently low to reduce thermal damage to substrate **122** and ensure adequate adhesion between substrate **122** and innermost layer **124**.

[0053] In some examples, computing device **12** may receive thermal data from at least one sensor during deposition of innermost layer **124**. The thermal data is indicative of a size and/or temperature of melt pool **126A**. Computing device **12** may further determine and/or adjust the first

set of deposition parameters based on the thermal data. For example, computing device **12** may control one or more deposition parameters related to the first amount of energy **128A** to maintain the temperature and/or size of melt pool **126A** below a target threshold. This target threshold may correspond to a temperature of substrate **122** deemed acceptable for maintaining a low or absent amount of thermal damage.

[0054] In some examples, computing device **12** may receive beam data from at least one sensor during deposition of innermost layer **124**. The beam data is indicative of an energy beam size from energy delivery device **16**. Computing device **12** may further determine and/or adjust the first set of deposition parameters based on the beam data. For example, computing device **12** may control one or more deposition parameters related to the first amount of energy **128A** to maintain the size of a beam of energy **128A** below a target threshold. This target threshold may correspond to a temperature of substrate **122** deemed acceptable for maintaining a low or absent amount of thermal damage.

[0055] The example method of FIG. 3 includes determining a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer (**104**). FIG. 4B is a cross-sectional side view diagram illustrating deposition of an inner plurality of layers **134** (“inner layers **134**”) of a deposit **120B** on innermost layer **124**. In the example of FIG. 4B, only a single, initial layer **134A** of inner layers **134** is shown. While inner layers **134** may not directly contact substrate **122** like innermost layer **124**, inner layers **134** may still have a greater effect on an integrity of deposit **120B** than other, overlying layers due to a proximity to substrate **122**. In some examples, a depth of inner layers **134** may be less than about 1 centimeter, such as less than about 1 millimeter.

[0056] The second set of deposition parameters is configured to cause an energy delivery device to deliver a second amount of energy **128B** to a build surface **123B** of deposit **120B** to form a melt pool **126B** in build surface **123B**, and cause a powder delivery device to direct a powder stream **129B** toward melt pool **126B**, thereby fusing the powder and forming innermost layer **124**. The second amount of energy **128B** may cause at least a portion of innermost layer **124** to melt.

[0057] The second set of deposition parameters may influence a quality of inner layers **134**. For example, a number and size of defects in inner layers **134** may increase with increasing powder flow rate of powder stream **129B** and increasing amount of energy **128B**. When subjected to stresses during operation of a component, these defects may form cracks or other more severe anomalies that negatively affect an integrity of deposit **120B**. For inner layer **134**, stresses may be particularly high due to a proximity of inner layers **134** to substrate **122**. For example, a coefficient of thermal expansion of deposit **120B** may be different from a coefficient of thermal expansion of substrate **122**, creating thermal stresses in response to a change in temperature. These thermal stresses may cause or accelerate cracks from the defects, resulting in delamination or other damage near an interface of substrate **122** and deposit **120B**.

[0058] In some examples, the second set of deposition parameters may be configured to limit the second amount of energy **128B** to build surface **123B** to reduce a number and/or size of defects in inner layers **134**. For example, the second set of deposition parameters may be configured to maintain a size and/or temperature of melt pool **126B** at or below a threshold that corresponds to reduced build quality of inner layers **134**. A smaller size and/or lower temperature of melt pool **126B** may generally result in more uniform deposition parameters at melt pool **126B** and, correspondingly, a reduced number and/or magnitude of defects in inner layers **134**. In some examples, a predicted build quality of inner layers **134** may be determined through materials modeling. For example, computing device **12** may model a predicted microstructure that results from a particular set of deposition parameters and select the second set of parameters that would reduce a number or magnitude of defects based on the modeling.

[0059] In addition to melting part of innermost layer **124**, energy **128B** may further cause a portion of innermost layer **124** and substrate **122** to heat above a temperate that may affect an integrity of

innermost layer **124** and/or substrate **122**, as indicated by heat affected zone **127B**. In examples in which deposit **120B** and substrate **122** are formed from different materials, substrate **122** may be particularly susceptible to thermal damage due to different thermal properties of substrate **122** and deposit **120B**. For example, a melting point and/or sintering temperature of deposit **120B** may be greater than a melting point of substrate **122**, such that outer portions of substrate **122** may melt prior to inner portions of innermost layer **124**. In some instances, a temperature of substrate **122** may increase above a melting point of substrate **122**, causing substrate **122** to re-melt (“melt back”).

[0060] In some examples, the second set of deposition parameters may be configured to limit the second amount of energy **128B** to build surface **123B** to reduce a temperature of substrate **122**. For example, the second set of deposition parameters may be configured to maintain a size and/or temperature of melt pool **126B** at or below a threshold that corresponds to melt back of substrate **122**. A smaller size and/or lower temperature of melt pool **126B** may generally result in a lower temperature of substrate **122** and, correspondingly, a reduced melt back of substrate **122**. The first set of deposition parameters may be configured to reduce a size of heat affected zone **127B** while still maintaining adequate energy **128B** for forming inner layers **134**.

[0061] In some examples, the second set of deposition parameters may be configured to permit cooling of substrate **122** to reduce a temperature of substrate **122**. For example, the first set of deposition parameters may define a toolpath of melt pool **126B** that permits portions of substrate **122** recently exposed to heat affected zone **127B** sufficient time to cool, such that a temperature of the particular portion may be reduced.

[0062] The example method of FIG. 3 includes depositing the inner plurality of layers on the innermost layer based on the second set of deposition parameters (**106**). For example, computing device **12** may control energy delivery device **16** and powder delivery device **14** to deposit the inner layers **134** on innermost layer **124** based on the second set of deposition parameters. During deposition, deposition conditions at melt pool **126A** may remain sufficiently constant to reduce defects in inner layers **134** and/or a temperature of substrate **122** may remain sufficiently low to prevent melt back to substrate **122**.

[0063] In some examples, computing device **12** may receive thermal data and/or beam data from at least one sensor during deposition of inner layers **134**. Computing device **12** may further determine and/or adjust the first set of deposition parameters based on the thermal and/or beam data. For example, computing device **12** may control one or more deposition parameters related to the second amount of energy **128B** to maintain the temperature and/or size of melt pool **126A** below a target threshold and/or maintain the size of a beam of energy **128A** below a target threshold. These target thresholds may correspond to a sufficiently low formation rate of inner layers **134** for reducing formation of defects and/or sufficiently low temperature of substrate **122** for avoiding melt back.

[0064] While innermost layer **124** and inner layers **134** have been described as having different target results, in some examples, the first set of deposition parameters is substantially similar to the second set of deposition parameters. For example, build quality of inner layers **134**, which may include innermost layer **124**, may be more limiting than a temperature effect of energies **128A** and **128B** to substrate **122**, such that deposition of both innermost layer **124** and inner layers **134** may be controlled according to a same or similar set of deposition parameters to reduce defects. In some examples, the first set of deposition parameters is different from the second set of deposition parameters. For example, substrate **122** may be more sensitive to thermal damage when depositing innermost layer **124** than melt back when depositing inner layers **134**, such that first amount of energy **128A** may be less than the second amount of energy **128B**.

[0065] The example method of FIG. 3 includes determining a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers (**108**). FIG. 4C is a cross-sectional side view diagram illustrating deposition of an outer plurality of layers **144** of a

deposit **120C** on inner layers **134**. The third set of deposition parameters is configured to cause an energy delivery device to deliver a third amount of energy **128C** to a build surface **123C** of deposit **120C** to form a melt pool **126C** in build surface **123C**, and cause a powder delivery device to direct a powder stream **129C** toward melt pool **126C**, thereby fusing the powder and forming outer layer **144A**. The third set of deposition parameters may be configured to maintain a size of melt pool **126C** to efficiently fabricate a remainder of component **22**. The third set of deposition parameters is different from the first and second sets of deposition parameters, and corresponds to a set of parameters for forming a bulk of a component that may be sufficiently removed from substrate **122** to not substantially affect adhesion of deposit **120C** to substrate **122**. As such, the third amount of energy **128C** is different than the first and second amounts of energy **128A** or **128B**. For example, outer layer **144A** may be formed at a faster rate than underlying layers, such that the third amount of energy may be relatively higher than for an otherwise slower rate. However, underlying layers of outer layer **144A** during deposition of outer layer **144A** may be at a relatively high temperature compared to substrate **122** during deposition of the underlying layers, such that the third amount of energy may be relatively lower than the second amount of energy, despite outer layer **144A** being formed at a faster rate. In some examples, the third amount of energy may be greater than about 10 percent and less than about 50 percent of each of the first amount of energy and the second amount of energy. Deposition parameters of the third set of deposition parameters that are different than the first and second sets of deposition parameters include at least one of move pause, a power and/or duty cycle of the energy delivery device, a thickness of the corresponding layer, a size of the melt pool, or a toolpath of the melt pool.

[0066] The example method of FIG. 3 includes controlling an energy delivery device and a powder delivery device to deposit outer plurality of layers on the inner plurality of layers based on third set of deposition parameters (**110**). For example, computing device **12** may control energy delivery device **16** and powder delivery device **14** to deposit the outer plurality of layers **144** on inner layers **134** based on third set of deposition parameters.

[0067] The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware, or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit including hardware may also perform one or more of the techniques of this disclosure.

[0068] Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various techniques described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware, firmware, or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware, firmware, or software components, or integrated within common or separate hardware, firmware, or software components.

[0069] The techniques described in this disclosure may also be embodied or encoded in an article of manufacture including a computer-readable storage medium encoded with instructions. Instructions embedded or encoded in an article of manufacture including a computer-readable storage medium encoded, may cause one or more programmable processors, or other processors, to implement one or more of the techniques described herein, such as when instructions included or encoded in the computer-readable storage medium are executed by the one or more processors.

Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a compact disc ROM (CD-ROM), a floppy disk, a cassette, magnetic media, optical media, or other computer readable media. In some examples, an article of manufacture may include one or more computer-readable storage media.

[0070] In some examples, a computer-readable storage medium may include a non-transitory medium. The term “non-transitory” may indicate that the storage medium is not embodied in a carrier wave or a propagated signal. In certain examples, a non-transitory storage medium may store data that can, over time, change (e.g., in RAM or cache).

[0071] Example 1: An additive manufacturing system includes an energy delivery device configured to deliver energy to a build surface of a deposit overlying a substrate to form a melt pool in the build surface; a powder delivery device configured to direct a powder stream toward the melt pool; and a computing device configured to: determine a first set of deposition parameters for an innermost layer of the deposit overlying the substrate; determine a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer; determine a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers, wherein the third set of deposition parameters is different from the first and second sets of deposition parameters; and control the energy delivery device and the powder delivery device to: deposit the innermost layer on the substrate based on the first set of deposition parameters, wherein, when operating based on the first set of deposition parameters, the energy delivery device delivers a first amount of energy to the build surface; deposit the inner plurality of layers on the innermost layer based on the second set of deposition parameters, wherein, when operating based on the second set of deposition parameters, the energy delivery device delivers a second amount of energy to the build surface; and deposit the outer plurality of layers on the inner plurality of layers based on third set of deposition parameters, wherein, when operating based on the third set of deposition parameters, the energy delivery device delivers a third amount of energy to the build surface, and wherein the third amount of energy is different than each of the first and second amounts of energy.

[0072] Example 2: The additive manufacturing system of example 1, wherein a melting point of the deposit is greater than a melting point of the substrate.

[0073] Example 3: The additive manufacturing system of any of examples 1 and 2, wherein the first set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to thermal damage of the substrate.

[0074] Example 4: The additive manufacturing system of any of examples 1 through 3, wherein the second set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to melt back of the substrate.

[0075] Example 5: The additive manufacturing system of any of examples 1 through 4, wherein deposition parameters of the third set of deposition parameters that are different from the first and second sets of deposition parameters include at least one of move pause, a power and/or duty cycle of the energy delivery device, a thickness of the corresponding layer, a size of the melt pool, or a toolpath of the melt pool.

[0076] Example 6: The additive manufacturing system of any of examples 1 through 5, further includes receive the thermal data from the at least one thermal sensor; and determine at least the first set of deposition parameters and the second set of deposition parameters based on the thermal data.

[0077] Example 7: The additive manufacturing system of any of examples 1 through 6, wherein a depth of the inner plurality of layers is less than about 1 centimeter.

[0078] Example 8: The additive manufacturing system of any of examples 1 through 7, wherein the first set of deposition parameters, the second set of deposition parameters, and the third set of

deposition parameters are substantially similar.

[0079] Example 9: The additive manufacturing system of any of examples 1 through 8, wherein the first set of deposition parameters is different from the second set of deposition parameters, and wherein the first amount of energy is greater than the second amount of energy.

[0080] Example 10: The additive manufacturing system of any of examples 1 through 9, wherein the third amount of energy is greater than about 10 percent and less than about 50 percent of each of the first amount of energy and the second amount of energy.

[0081] Example 11: A method for additive manufacturing includes determining, by a computing device, a first set of deposition parameters for an innermost layer of a deposit overlying a substrate; determining, by the computing device, a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer; determining, by the computing device, a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers, wherein the third set of deposition parameters is different from the first and second sets of deposition parameters; and controlling, by the computing device, an energy delivery device to deliver energy to the build surface to form a melt pool and a powder delivery device to direct a powder stream toward the melt pool to: deposit the innermost layer on the substrate based on the first set of deposition parameters, wherein, when operating based on the first set of deposition parameters, the energy delivery device delivers a first amount of energy to the build surface; deposit the inner plurality of layers on the innermost layer based on the second set of deposition parameters, wherein, when operating based on the second set of deposition parameters, the energy delivery device delivers a second amount of energy to the build surface; and deposit the outer plurality of layers on the inner plurality of layers based on third set of deposition parameters, wherein, when operating based on the third set of deposition parameters, the energy delivery device delivers a third amount of energy to the build surface, and wherein the third amount of energy is different than each of the first and second amounts of energy.

[0082] Example 12: The method of example 11, wherein a melting point of the deposit is greater than a melting point of the substrate.

[0083] Example 13: The method of any of examples 11 and 12, wherein the first set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to thermal damage of the substrate.

[0084] Example 14: The method of any of examples 11 through 13, wherein the second set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to melt back of the substrate.

[0085] Example 15: The method of any of examples 11 through 14, wherein deposition parameters of the third set of deposition parameters that are different from the first and second sets of deposition parameters include at least one of move pause, a power and/or duty cycle of the energy delivery device, a thickness of the corresponding layer, a size of the melt pool, or a toolpath of the melt pool.

[0086] Example 16: The method of any of examples 11 through 15, further includes receiving, by the computing device, thermal data from at least one thermal sensor, wherein the thermal data is indicative of a temperature of the melt pool; and determining, by the computing device, at least the first set of deposition parameters and the second set of thermal conditions based on the thermal data.

[0087] Example 17: The method of any of examples 11 through 16, wherein a depth of the inner plurality of layers is less than about 1 centimeter.

[0088] Example 18: The method of any of examples 11 through 17, wherein the first set of deposition parameters, the second set of deposition parameters, and the third set of deposition parameters are substantially similar.

[0089] Example 19: The method of any of examples 11 through 18, wherein the first set of deposition parameters is different from the second set of deposition parameters, and wherein the

first amount of energy is greater than the second amount of energy.

[0090] Example 20: The method of any of examples 11 through 19, wherein the third amount of energy is greater than about 10 percent and less than about 50 percent of each of the first amount of energy and the second amount of energy.

[0091] Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. An additive manufacturing system, comprising: an energy delivery device configured to deliver energy to a build surface of a deposit overlying a substrate to form a melt pool in the build surface; a powder delivery device configured to direct a powder stream toward the melt pool; and a computing device configured to: determine a first set of deposition parameters for an innermost layer of the deposit overlying the substrate; determine a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer; determine a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers, wherein the third set of deposition parameters is different from the first and second sets of deposition parameters; and control the energy delivery device and the powder delivery device to: deposit the innermost layer on the substrate based on the first set of deposition parameters, wherein, when operating based on the first set of deposition parameters, the energy delivery device delivers a first amount of energy to the build surface; deposit the inner plurality of layers on the innermost layer based on the second set of deposition parameters, wherein, when operating based on the second set of deposition parameters, the energy delivery device delivers a second amount of energy to the build surface; and deposit the outer plurality of layers on the inner plurality of layers based on third set of deposition parameters, wherein, when operating based on the third set of deposition parameters, the energy delivery device delivers a third amount of energy to the build surface, and wherein the third amount of energy is different than each of the first and second amounts of energy.
2. The additive manufacturing system of claim 1, wherein a melting point of the deposit is greater than a melting point of the substrate.
3. The additive manufacturing system of claim 1, wherein the first set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to thermal damage of the substrate.
4. The additive manufacturing system of claim 1, wherein the second set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to melt back of the substrate.
5. The additive manufacturing system of claim 1, wherein deposition parameters of the third set of deposition parameters that are different from the first and second sets of deposition parameters include at least one of move pause, a power and/or duty cycle of the energy delivery device, a thickness of the corresponding layer, a size of the melt pool, or a toolpath of the melt pool.
6. The additive manufacturing system of claim 1, further comprising at least one thermal sensor configured to detect thermal data indicative of a temperature of the melt pool, and wherein the computing device is further configured to: receive the thermal data from the at least one thermal sensor; and determine at least the first set of deposition parameters and the second set of deposition parameters based on the thermal data.
7. The additive manufacturing system of claim 1, wherein a depth of the inner plurality of layers is less than about 1 centimeter.
8. The additive manufacturing system of claim 1, wherein the first set of deposition parameters, the second set of deposition parameters, and the third set of deposition parameters are substantially similar.

9. The additive manufacturing system of claim 1, wherein the first set of deposition parameters is different from the second set of deposition parameters, and wherein the first amount of energy is greater than the second amount of energy.

10. The additive manufacturing system of claim 1, wherein the third amount of energy is greater than about 10 percent and less than about 50 percent of each of the first amount of energy and the second amount of energy.

11. A method for additive manufacturing, comprising: determining, by a computing device, a first set of deposition parameters for an innermost layer of a deposit overlying a substrate; determining, by the computing device, a second set of deposition parameters for an inner plurality of layers of the deposit overlying the innermost layer; determining, by the computing device, a third set of deposition parameters for an outer plurality of layers of the deposit overlying the inner plurality of layers, wherein the third set of deposition parameters is different from the first and second sets of deposition parameters; and controlling, by the computing device, an energy delivery device to deliver energy to the build surface to form a melt pool and a powder delivery device to direct a powder stream toward the melt pool to: deposit the innermost layer on the substrate based on the first set of deposition parameters, wherein, when operating based on the first set of deposition parameters, the energy delivery device delivers a first amount of energy to the build surface; deposit the inner plurality of layers on the innermost layer based on the second set of deposition parameters, wherein, when operating based on the second set of deposition parameters, the energy delivery device delivers a second amount of energy to the build surface; and deposit the outer plurality of layers on the inner plurality of layers based on third set of deposition parameters, wherein, when operating based on the third set of deposition parameters, the energy delivery device delivers a third amount of energy to the build surface, and wherein the third amount of energy is different than each of the first and second amounts of energy.

12. The method of claim 11, wherein a melting point of the deposit is greater than a melting point of the substrate.

13. The method of claim 11, wherein the first set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to thermal damage of the substrate.

14. The method of claim 11, wherein the second set of deposition parameters is configured to maintain a size of the melt pool at or below a threshold that corresponds to melt back of the substrate.

15. The method of claim 11, wherein deposition parameters of the third set of deposition parameters that are different from the first and second sets of deposition parameters include at least one of move pause, a power and/or duty cycle of the energy delivery device, a thickness of the corresponding layer, a size of the melt pool, or a toolpath of the melt pool.

16. The method of claim 11, further comprising: receiving, by the computing device, thermal data from at least one thermal sensor, wherein the thermal data is indicative of a temperature of the melt pool; and determining, by the computing device, at least the first set of deposition parameters and the second set of thermal conditions based on the thermal data.

17. The method of claim 11, wherein a depth of the inner plurality of layers is less than about 1 centimeter.

18. The method of claim 11, wherein the first set of deposition parameters, the second set of deposition parameters, and the third set of deposition parameters are substantially similar.

19. The method of claim 11, wherein the first set of deposition parameters is different from the second set of deposition parameters, and wherein the first amount of energy is greater than the second amount of energy.

20. The method of claim 11, wherein the third amount of energy is greater than about 10 percent and less than about 50 percent of each of the first amount of energy and the second amount of energy.
